Department of Computer Science and Engineering

PROJECT REPORT

Project Title:

Local Disaster Relief and Volunteer Coordination Database

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Department of CSE, IIUC.

Local Disaster Relief and Volunteer Coordination Database

1. Introduction:

The Local Disaster Relief and Volunteer Coordination Database project is designed to streamline and centralize disaster relief efforts. It focuses on efficiently managing volunteers, tracking resources like food and medical supplies, and coordinating shelter capacities during emergencies. The system registers volunteers, tracks their skills and availability, handles requests from disaster zones, and prioritizes resource allocation based on the severity of situations. By leveraging data analytics and real-time updates, the project aims to enhance disaster response efficiency, ensuring that help reaches where it's needed most on time.

2. Literature Review:

Existing works and URL	Findings	Effects
American Red Cross Volunteer Management System. **Link:** https://www.redcross.org/	Complex, Hard to adapt quickly with local disasters, needs extensive training	Lacks real-time updates on shelter availability and resource allocation for local relief efforts.
UN Volunteers (UNV) Platform **Link:** https://app.unv.org/	Lacks functionality for managing local shelters and tracking resource needs. Coordination is often slow	Its global approach makes it inefficient for immediate, localized disaster situations.
Crisis Cleanup **Link:** https://crisiscleanup.org/live	It does not provide proactive tracking of shelter capacities or real-time resource management.	The platform neglects pre-disaster preparedness and relies on manual updates, delaying responses.

Effectiveness of Our Project:

The Local Disaster Relief and Volunteer Coordination Database overcomes these limitations by:

- Streamlining Local Coordination: Real-time updates for volunteers, shelters, and resources.
- Proactive Resource Management: Tracks shortages to prevent critical gaps during emergencies.
- Simplified Interface: Tailored for quick use in local disaster scenarios without complex overhead.

Our project provides a targeted, efficient, and real-time solution for small-scale, localized disaster relief efforts, addressing gaps that existing platforms overlook.

3.Aims and Objective:

1. Register and track volunteers, including their skills and availability, to ensure they are effectively assigned to tasks during emergencies.

2. Manage and track critical resources such as food, medical supplies, and shelter capacities, prioritizing allocation based on the severity of disaster situations.

3. Centralize disaster relief efforts by streamlining communication and coordination among volunteers, shelters, and relief agencies.

4. Handle real-time requests from disaster zones, such as medical aid or rescue operations, and prioritize responses based on urgency.

5. Leverage data analytics to generate real-time updates and reports, improving the efficiency and effectiveness of disaster relief operations.

SQL Queries and Output with Description

1.Show All Table Data with Related Data :

This query selects all columns from multiple tables,joining them on their related keys to provide a comprehensive view of all related data.

SQL QUERIES:

```sql
SELECT
    V.VolunteerID, V.Name AS VolunteerName, V.Skills, V.AvailabilityStatus, V.CertificationStatus,
    RG.Name AS RegionName, RG.Population,
    A.AssignmentID, A.AssignmentStatus,
    R.SeverityLevel, R.ResourceType, R.QuantityNeeded,
    S.Name AS ShelterName, S.TotalCapacity, S.AvailableCapacity, S.ResourceAvailabilityStatus,
    RE.ResourceType, RE.QuantityAvailable,
    DZ.DisasterType, DZ.ActiveStatus,
    BR.SeverityLevel AS BackupSeverity, BR.ResourceType AS BackupResource, BR.QuantityNeeded AS BackupQuantity
FROM Volunteers V
LEFT JOIN Assignments A ON V.VolunteerID = A.VolunteerID
LEFT JOIN Requests R ON A.RequestID = R.RequestID
LEFT JOIN Regions RG ON V.RegionID = RG.RegionID
LEFT JOIN Shelters S ON RG.RegionID = S.RegionID
LEFT JOIN Resources RE ON S.ShelterID = RE.ShelterID
LEFT JOIN DisasterZones DZ ON RG.RegionID = DZ.RegionID
LEFT JOIN BackupRequests BR ON R.RequestID = BR.RequestID;
```

OUTPUT:

Results Messages

	VolunteerID	VolunteerName	Skills	AvailabilityStatus	CertificationStatus	RegionName	Population	AssignmentID	AssignmentStatus	SeverityLevel	ResourceType	QuantityNeeded
1	1	Ahnaf Hossain	Medical	Available	Certified	Dhaka	8900000	1	Completed	Critical	Food	100
2	2	Ibrahim Abdullah	Logistics	Unavailable	Certified	Chittagong	2600000	2	Pending	High	Medical Supplies	50
3	3	Ali Khan	Rescue Operations	Available	Pending	Sylhet	1500000	3	Ongoing	Medium	Water	200
4	4	Maria Ahmed	Medical, Rescue	Available	Certified	Khulna	1400000	4	Pending	Critical	Shelter	20
5	5	Rafiq Islam	Food Distribution	Unavailable	Certified	Rajshahi	1250000	5	Completed	Low	Clothing	80

ShelterName	TotalCapacity	AvailableCapacity	ResourceAvailabilityStatus	ResourceType	QuantityAvailable	DisasterType	ActiveStatus	BackupSeverity	BackupResource	BackupQuantity
Shelter A	500	450	Adequate	Food	300	Flood	Active	Critical	Food	100
Shelter B	300	100	Critical	Water	500	Cyclone	Active	High	Medical Supplies	50
Shelter C	200	200	Adequate	Medical Supplies	200	Earthquake	Inactive	Medium	Water	200
Shelter D	400	50	Critical	Blankets	150	Landslide	Inactive	Critical	Shelter	20
Shelter E	350	150	Moderate	Clothing	100	Drought	Active	Low	Clothing	80

2.Update Table Data:

This update query change the volunteers name according to their ID

SQL QUERIES:

```
UPDATE Volunteers SET Name = 'Ahnaf Hossain' WHERE VolunteerID = 1;

UPDATE Volunteers SET Name = 'Ibrahim Abdullah' WHERE VolunteerID = 2;

SELECT * FROM Volunteers
```

OUTPUT:

75 %

Results Messages

	VolunteerID	Name	Skills	AvailabilityStatus	CertificationStatus	RegionID
1	1	Ahnaf Hossain	Medical	Available	Certified	1
2	2	Ibrahim Abdullah	Logistics	Unavailable	Certified	2
3	3	Ali Khan	Rescue Operations	Available	Pending	3
4	4	Maria Ahmed	Medical, Rescue	Available	Certified	4
5	5	Rafiq Islam	Food Distribution	Unavailable	Certified	5

3.Inserting Data in Table:

We can insert various data in various table

SQL QUERIES:

```sql
INSERT INTO Volunteers (Name, Skills, AvailabilityStatus, CertificationStatus, RegionID)
VALUES ('Lucas  Velverde', 'General Assistance', 'Available', 'Certified', 1);

SELECT * FROM Volunteers
```

OUTPUT:

5 %

Results | Messages

	VolunteerID	Name	Skills	AvailabilityStatus	CertificationStatus	RegionID
1	1	Ahnaf Hossain	Medical	Available	Certified	1
2	2	Ibrahim Abdullah	Logistics	Unavailable	Certified	2
3	3	Ali Khan	Rescue Operations	Available	Pending	3
4	4	Maria Ahmed	Medical, Rescue	Available	Certified	4
5	5	Rafiq Islam	Food Distribution	Unavailable	Certified	5
6	6	Lucas Velverde	General Assistance	Available	Certified	1

4.Delete Data from Table:

We can delete unwanted data from various table

SQL QUERIES:

```sql
DELETE FROM Volunteers WHERE Name = 'Lucas  Velverde';

SELECT * FROM Volunteers
```

OUTPUT:

Results | Messages

	VolunteerID	Name	Skills	AvailabilityStatus	CertificationStatus	RegionID
1	1	Ahnaf Hossain	Medical	Available	Certified	1
2	2	Ibrahim Abdullah	Logistics	Unavailable	Certified	2
3	3	Ali Khan	Rescue Operations	Available	Pending	3
4	4	Maria Ahmed	Medical, Rescue	Available	Certified	4
5	5	Rafiq Islam	Food Distribution	Unavailable	Certified	5

5. Order By Searching:

This query sorts the table in a descending way by the quantity needed

SQL QUERIES:

```
SELECT * FROM Requests
ORDER BY QuantityNeeded DESC;
```

OUTPUT:

RequestID	RegionID	SeverityLevel	ResourceType	QuantityNeeded
3	3	Medium	Water	200
1	1	Critical	Food	100
5	5	Low	Clothing	80
2	2	High	Medical Supplies	50
4	4	Critical	Shelter	20

6. Min And Max Function:

Using these queries, we can find out the maximum and minimum shelter capacity

SQL QUERIES:

```
SELECT *
FROM Shelters
WHERE AvailableCapacity = (SELECT MIN(AvailableCapacity) FROM Shelters);
```

```
SELECT *
FROM Shelters
WHERE AvailableCapacity = (SELECT MAX(AvailableCapacity) FROM Shelters);
```

OUTPUT:

<u>Min</u>:

ShelterID	Name	RegionID	TotalCapacity	AvailableCapacity	ResourceAvailabilityStatus
4	Shelter D	4	400	50	Critical

<u>Max</u>:

ShelterID	Name	RegionID	TotalCapacity	AvailableCapacity	ResourceAvailabilityStatus
1	Shelter A	1	500	450	Adequate

7.Count Function:

We can count how many regions have critical situations

SQL QUERIES:

```
SELECT COUNT(*) AS CriticalRequests
FROM Requests
WHERE SeverityLevel = 'Critical';
```

OUTPUT:

CriticalRequests
2

8.Sum Function:

We can sum the total resource available in all shelters combined

SQL QUERIES:

```
SELECT SUM(QuantityAvailable) AS TotalResources
FROM Resources;
```

OUTPUT:

TotalResources
1250

9. Like Wise Search:

Those who have medical skill we can get their data

SQL QUERIES:

```sql
SELECT * FROM Volunteers
WHERE Skills LIKE '%Medical%';
```

OUTPUT:

VolunteerID	Name	Skills	AvailabilityStatus	CertificationStatus	RegionID
1	Ahnaf Hossain	Medical	Available	Certified	1
4	Maria Ahmed	Medical, Rescue	Available	Certified	4

10. Union Function:

We can collect the data of volunteers and their regions from two table region and volunteers

SQL QUERIES:

```
SELECT V.VolunteerID, V.Name, R.Name AS Region
FROM Volunteers V
JOIN Regions R ON V.RegionID = R.RegionID
WHERE R.Name = 'Dhaka'

UNION

SELECT V.VolunteerID, V.Name, R.Name AS Region
FROM Volunteers V
JOIN Regions R ON V.RegionID = R.RegionID
WHERE R.Name = 'Chittagong';
```

OUTPUT:

VolunteerID	Name	Region
1	Ahnaf Hossain	Dhaka
2	Ibrahim Abdullah	Chittagong

11. Distinct Function:

It gives us distinct resources

SQL QUERIES:

```
SELECT DISTINCT ResourceType
FROM Resources;
```

OUTPUT:

ResourceType
Blankets
Clothing
Food
Medical Supplies
Water

12. Case Statement:

The CASE statement in SQL is used to perform conditional logic and return specific values based on conditions.

SQL QUERIES:

```
SELECT Name,
CASE
    WHEN AvailabilityStatus = 'Available' THEN 'Active'
    ELSE 'Inactive'
END AS AvailabilityCategory
FROM Volunteers;
```

OUTPUT:

	Name	AvailabilityCategory
1	Ahnaf Hossain	Active
2	Ibrahim Abdullah	Inactive
3	Ali Khan	Active
4	Maria Ahmed	Active
5	Rafiq Islam	Inactive

13. Having Clause:

The HAVING clause filters grouped data based on conditions after applying aggregate functions.

SQL QUERIES:

```
SELECT ShelterID, AvailableCapacity
FROM Shelters
GROUP BY ShelterID, AvailableCapacity
HAVING AvailableCapacity < 200;
```

OUTPUT:

ShelterID	AvailableCapacity
2	100
4	50
5	150

14.Conditional Query:

It searches according to the condition

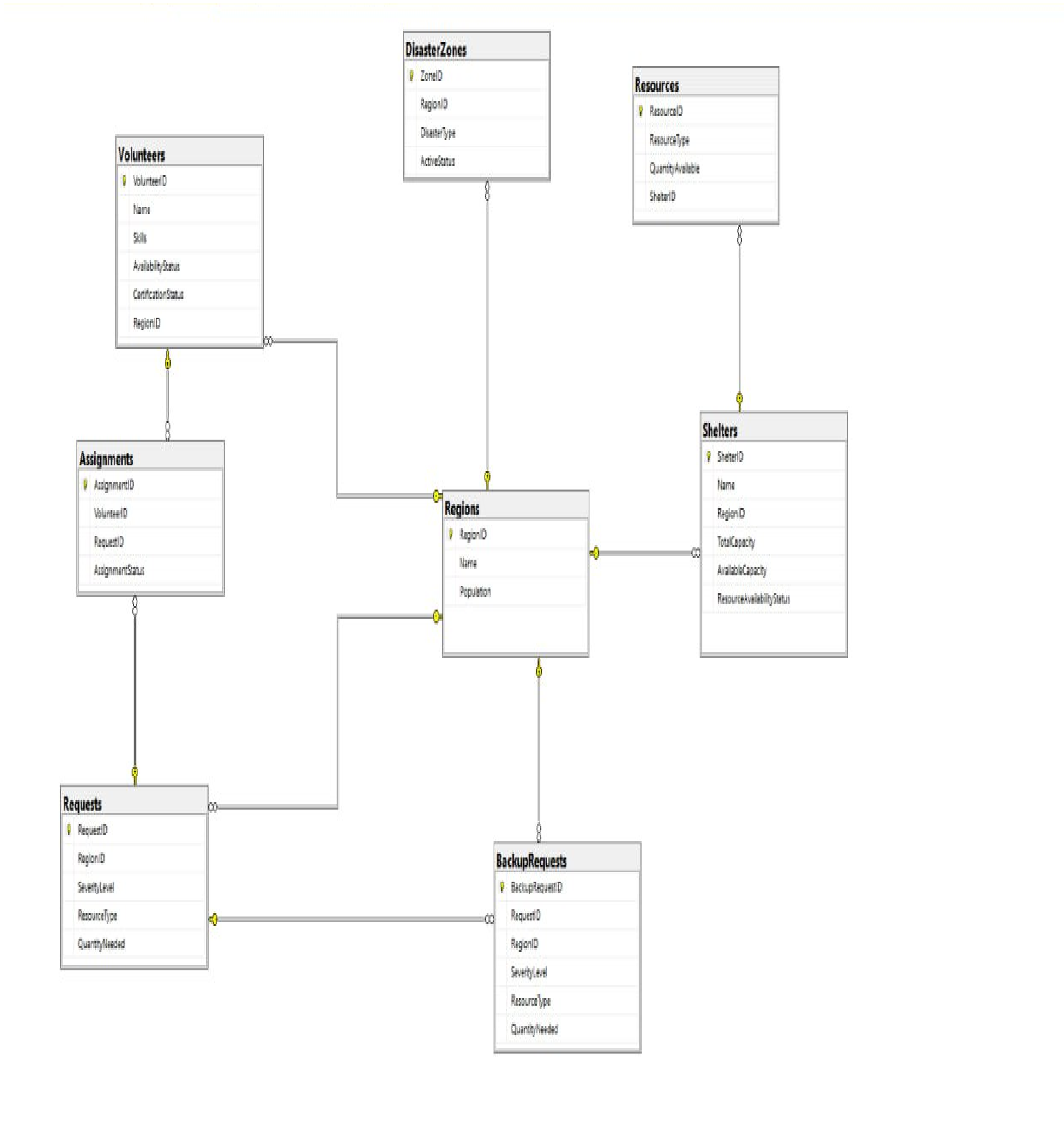
SQL QUERIES:

```
SELECT * FROM Requests
WHERE SeverityLevel = 'Critical';
```

OUTPUT:

RequestID	RegionID	SeverityLevel	ResourceType	QuantityNeeded
1	1	Critical	Food	100
4	4	Critical	Shelter	20

DATABASE DIAGRAM



Conclusion:

The **Local Disaster Relief and Volunteer Coordination Database** project aims to bridge critical gaps in disaster management by leveraging technology to streamline relief efforts and enhance volunteer coordination. By providing a centralized platform, the system ensures efficient resource allocation, real-time tracking of shelters, and effective communication among stakeholders. Its ability to manage and match volunteers based on skills and availability, combined with tools for handling requests from disaster zones, ensures that relief efforts are both targeted and impactful.

Moreover, the incorporation of data security measures safeguards sensitive information, fostering trust among users. Real-time analytics and reporting functionalities allow stakeholders to assess the effectiveness of disaster responses and make informed decisions during crises.

In conclusion, this project provides a robust and scalable solution to modern disaster management challenges, ensuring timely and organized responses to emergencies. It empowers relief organizations, volunteers, and affected communities by creating a collaborative and efficient framework that minimizes delays and maximizes the impact of disaster relief efforts.

References:

1.W3School SQL

Link: https://www.w3schools.com/sql/default.asp

2. American Red Cross Volunteer Management System.

Link: https://www.redcross.org/

3. UN Volunteers (UNV) Platform

Link: https://app.unv.org/

4. Crisis Cleanup
Link: https://crisiscleanup.org/live